

Integrity Framework: Security vs Performance Trade-off

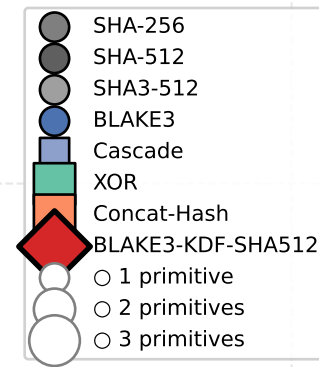
Quantum Security (bits)

256

192

128

↗ Upper-right is better



XOR

BLAKE3-KDF-SHA512

BLAKE3-SHA512

Cascade

SHA-512

BLAKE3

SHA-256

128-bit threshold

Integrity Throughput at 1 KB (MB/s)

200

400

600

800

1000